UNIT 9 · TOPIC 5(B)

Measurement of Force, Torque, Power, Sound and Vibration

Syllabus topic: Measurement of temperature, pressure, strain, force, torque, power, sound level and vibrations

Force, torque and power

Quantity	Instruments
FORCE	Strain-gauge LOAD CELL (the standard for draft/weight — Topic 2); PROVING RING (calibrated elastic ring with dial); spring balance; hydraulic load cells
TORQUE	Strain-gauged torsion shaft or torsion bar (angular twist ∝ torque), read from a rotating shaft through slip rings/telemetry
POWER	DYNAMOMETERS (Unit 3, Topic 12a): absorption types (rope brake, hydraulic, eddy-current, DC) and transmission types (torsion meter); PTO and drawbar dynamometers for tractors

$$\text{Brake power BP (kW)} = 2\pi NT/60{,}000;\ T\ (\text{N·m}) = 9550 \cdot P/N\ \text{(Unit 4, Topic 3a)}$$

Sound

- A **SOUND LEVEL METER** (microphone → amplifier → weighting network → display) reads sound pressure level in dB — usually A-weighted, dB(A), to match the ear.
- Octave-band and FFT analysers give the frequency content.
- Used for operator-noise and machine-noise assessment (OSHA 90 dB(A) for 8 h — Unit 5, Topic 6a).

$$SPL = 20 \cdot \log(p/p_0),\ p_0 = 20\ \mu Pa$$

Vibration

- A piezoelectric **ACCELEROMETER** (charge/voltage ∝ acceleration) is the standard pickup.
- Vibration is expressed as displacement, velocity or acceleration (RMS or peak) against frequency; an FFT/spectrum analyser resolves the frequency components.
- **Applications:** whole-body and hand-arm vibration (Unit 5, Topic 6b; ISO 2631/5349), machine condition monitoring, dynamic-balance testing (Unit 2, Topic 4).

Practice MCQs

Q47. Force is most commonly measured with a:

(a) Thermocouple (b) Strain-gauge load cell
(c) Rotameter (d) Pyrometer

Q48. A proving ring measures:

(a) Temperature (b) Force (elastic deflection)
(c) Flow (d) Sound

Q49. Torque on a rotating shaft is measured with:

(a) A manometer
(b) A strain-gauged torsion shaft (with slip rings/telemetry)
(c) A thermistor
(d) An orifice

Q50. Brake power from torque T and speed N (rpm) is:

(a) $2\pi NT/60000$
(b) $NT/60$
(c) T/N
(d) $9550NT$

Q51. Sound Pressure Level is defined as:

(a) $10 \log(p/p_0)$
(b) $20 \log(p/p_0)$
(c) $p - p_0$
(d) p/p_0

Q52. The reference pressure for SPL is:

(a) 1 Pa
(b) 20 μPa
(c) 1 bar
(d) 101 kPa

Q53. The A-weighting network in a sound level meter is used to:

(a) Amplify bass
(b) Match the response of the human ear (dBA)
(c) Measure flow
(d) Filter light

Q54. Vibration is most commonly sensed by a:

(a) Thermocouple
(b) Piezoelectric accelerometer
(c) Bourdon tube
(d) Rotameter

Q55. Frequency content of a vibration signal is obtained using:

(a) A manometer
(b) An FFT/spectrum analyser
(c) A calorimeter
(d) A viscometer

Answers

Q47 — (b) Strain-gauge load cell
Q48 — (b) Force (elastic deflection)
Q49 — (b) A strain-gauged torsion shaft (with slip rings/telemetry)
Q50 — (a) $2\pi NT/60000$ — *BP = 2πNT/60,000 (Unit 3, Topic 12a; Unit 4, Topic 3a).*
Q51 — (b) $20 \log(p/p_0)$ — *SPL = 20 log(p/p₀).*
Q52 — (b) 20 μPa — *p_0 = 20 μPa.*
Q53 — (b) Match the response of the human ear (dBA) — *A-weighting mimics the ear (Unit 5, Topic 6a).*
Q54 — (b) Piezoelectric accelerometer
Q55 — (b) An FFT/spectrum analyser

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney; measuring-instruments course notes (as given in the source notes, July 2026); standard fuel-testing methods (ASTM/IS); NPTEL measurement lectures.